MUHAMMAD HASSAN

AI/ML Engineer — Deep Learning,NLP, LLM & RAG Systems (LangChain) | Python Developer

Sargodha, Punjab, Pakistan | [03177498727] | [itxhassan6470@gmail.com] | [linkedin.com/in/muhammad-hassan-201a37323] [https://github.com/itxhassan6470-coder]

PROFESSIONAL SUMMARY

Final-year BS Information Technology student with hands-on experience across the machine learning and deep learning pipeline – from data preprocessing and model development to building deployable, security- and biometrics-focused applications. Built a real-time Intrusion Detection System using a Random Forest classifier with a live monitoring dashboard, a Facial Biometric Authentication System combining face recognition, liveness detection, and an anti-spoofing deep learning model, and a Retrieval-Augmented Generation (RAG) YouTube video Q&A chatbot using LangChain, vector embeddings, and LLMs. Comfortable working across Windows and Linux/VirtualBox environments, with practical experience in networking, virtualization, and IT infrastructure concepts that complement AI system design. Currently expanding into LangGraph and agentic AI systems. Seeking an AI/ML Engineer, Machine Learning Engineer, or Python Developer internship/junior role.

TECHNICAL SKILLS

Languages: Python, SQL,Html,css,javascript

ML / DL: scikit-learn, Random Forest, TensorFlow/Keras, CNN, model evaluation, feature engineering

Computer Vision: OpenCV, DeepFace, dlib, YOLO (YOLOv12n), Caffe SSD (face detection), liveness/anti-spoofing detection

Networking & Security: Intrusion Detection Systems, packet capture & analysis (Scapy), network protocols, CICIDS2017 dataset, virtualization (VirtualBox, multi-VM test environments)

Web / App Frameworks Streamlit

Databases: SQLite

Tools & Platforms: Git/GitHub, Windows, Linux (Ubuntu), VirtualBox,

PROJECTS

Real-Time Intrusion Detection System (IDS) — Final Year Project

Python, Scikit-learn (Random Forest), Flask, Socket.IO, Scapy, CICIDS2017

- Designed and trained a Random Forest classifier on the CICIDS2017 network traffic dataset to detect malicious activity in real time.
- Built a live monitoring dashboard using Flask and Socket.IO to stream detection alerts and traffic statistics to the browser as they occur.
- Implemented live packet capture and feature extraction using Scapy to feed real network traffic into the trained model.
- Set up a 2-VM attack simulation environment (dedicated IDS VM and Attacker VM) to test detection accuracy against simulated intrusion attempts.

Facial Biometric Authentication System with Liveness Detection

Python, OpenCV, DeepFace, dlib, PyTorch, Streamlit, SQLite, YOLOv12n

- Built an end-to-end biometric authentication pipeline combining face recognition, blink-based liveness detection, and a YOLO-based anti-spoofing model to classify real vs. fake (spoofed) faces.
- Implemented face detection using a Caffe SSD model and face recognition/embedding generation using DeepFace and dlib.
- Designed a relational schema (users, embeddings, audit_logs) in SQLite to store identity data and authentication history securely.
- Developed an interactive Streamlit front end for user enrollment and real-time authentication, built module-by-module (detection → recognition → anti-spoofing → audit logging).

YouTube Video Q&A Chatbot — Retrieval-Augmented Generation (RAG) System
Python, LangChain, Streamlit, FAISS, OpenAI/Google Gemini, YouTube Transcript API

Python, LangChain, Streamlit, FAISS, RAG,Google Gemini, YouTube Transcript API

- Built an end-to-end RAG pipeline that extracts YouTube video transcripts and enables natural-language Q&A grounded strictly in video content, preventing hallucinated answers.
- Implemented a modular retrieval chain using LangChain – text chunking (RecursiveCharacterTextSplitter), embedding generation, and similarity-based retrieval via a FAISS vector store.
- Designed an interactive Streamlit chat interface supporting session-based indexing, multi-turn conversation, and configurable retrieval parameters (chunk count, model selection).

EDUCATION

Bachelor of Science in Information Technology (BSIT) [2022] - [2026]
[GAMC], [SARGODHA]

RELEVANT COURSEWORK

Mobile & Wireless Communication (cellular networks, WLAN security & protocols) · IT Infrastructure (storage, compute, operating systems) · Virtual Systems & Services (virtualization architectures) · Networking · Machine Learning / Deep Learning

CERTIFICATIONS

Microsoft azure AI fundamentals

EXPERIENCE

AI/ML Engineer — Independent Projects

1 Year of Hands-on Experience

- Building expertise in **AI/ML, LLMs, RAG, NLP, Computer Vision, and Deep Learning**.
- Developed projects using **Python, LangChain, LLM APIs, FAISS, PyTorch, TensorFlow/Keras, OpenCV, and Scikit-learn**.
- Built and deployed AI applications using **Flask and Streamlit**, including RAG, real-time ML, and computer vision systems.